Interaction Floor Test

Does your bias audit measure people, or words?

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September 2026

The problem

An audit that asks whether two characteristics together hurt a person more than each one alone compares four versions of the same case: neither characteristic, the first, the second, and both. The interaction is

$$d = (\text{both} - \text{first}) - (\text{second} - \text{neither}).$$

That number can come out large and highly significant for reasons that have nothing to do with people. Many instruments that read a language model's probabilities respond to the *combination* of any two pieces of text, including two that mean nothing socially. Such an instrument reports a compounding effect for every pair it is given, and the audit reads it as bias.

The test

Alongside the pairs being audited, build FLOOR pairs: two socially neutral but distinctive details, inserted in exactly the same positions and the same way. "I cycle to work" with "I bake my own bread". "I walk my dog" with "I am learning the guitar". Their interaction should be zero.

If a floor pair's interaction is significant, the instrument responds to combining text, and none of the social interactions it reports can be trusted, however significant they are. One moving floor is enough: the verdict does not get to pick the floor it likes.

Use

```
python floor_test.py results.csv
python floor_test.py results.csv --json report.json
```

Python 3.8 or later, standard library only, no dependencies.

exit code	meaning
0	floor clean: the audited interactions can be read
2	floor contaminated, overall or within a stratum
1	the input could not be used

Input. One row per pair and item:

```
pair,kind,item,none,a,b,both
```

`pair` names the pair; `kind` is `social` for a pair being audited or `floor` for a neutral one; `item` is the unit the four versions share, such as a task, a profile or a question; the last four columns are the outcome in each version, on one scale. Zero or one, a score, a log probability: anything, as long as all four cells use the same scale. Use at least three floor pairs; the tool warns below that, because one neutral pair can itself be unusual.

Reading the verdict

Every pair is tested twice: by signed rank, and by a sign-flip permutation test on the mean, which is the number reported. On outcomes with a ceiling and a heavy tail the two disagree, and choosing between them after seeing the data is a forking path. So the rule only ever takes claims away: a neutral pair moves if EITHER test rejects, and an audited pair counts only if BOTH do and the mean leans the same way as the signed ranks.

CLEAN. Every neutral pair is quiet, overall and within thirds of the items ordered by difficulty, under one family-wise correction for the whole question: Holm across every floor test, then Bonferroni across the two tests. The audited interactions can be read, with the minimum detectable effect beside each one, because an absence of evidence smaller than that number is not evidence of absence.

CONTAMINATED. At least one neutral pair interacts. The instrument responds to combining text, and no audited interaction it reports can be treated as a finding. Nothing in the arithmetic repairs this; the fix is a different instrument.

CLEAN IN AGGREGATE ONLY. The neutral pairs are quiet on average but at least one moves within a difficulty stratum, typically one way on easy items and the other way on hard ones. The average hides it; the instrument is contaminated all the same.

What a sign means. A negative interaction is compounding harm only when both single characteristics are costs. When both help, a negative interaction means the help does not add. The report labels every audited interaction by the direction of its two single effects.

Designing an audit this test can check

Each rule was learned by getting it wrong first, and each one changed a conclusion.

1. **Hold the number of statements constant**, not only the length. A characteristic takes the place of a neutral statement; it never adds one. Otherwise "two characteristics" also means "one more sentence".
2. **Match length by whole statements**, never mid-phrase. A 40-word preamble moved one model's accuracy by ten points on its own.
3. **Put each characteristic in the same position** in the single and the combined version.
4. **Run both orders** wherever order can vary.
5. **Build at least three floor pairs.**

6. **Neutral means no cue** about age, disability, health, language, nationality, religion, family, gender, class or education. Check the words, not only the meaning: "reads crime novels" is a harmless hobby that contains the word "crime", and an instrument that reads text reads words.
7. **Distinctive means real information.** Filler is padding, not a floor.
8. **Keep outcomes paired within the item.**
9. **Never build an interaction by differencing margins** measured against a shared reference. Compare the combined version directly against each single instead, and put the floor through the same comparison.
10. **Check the instrument produces what you think it reads.** If you score the probability of "A" against "B", record how much probability the model put on those two letters at all.
11. **Collect the reference twice** if the study runs in more than one session, and check the two copies agree.
12. **Read the floor first**, before any audited result.
13. **Correct floors and audited pairs as two separate families.** Correcting them together makes a quiet floor easier to flag the more the audit finds.
14. **Report the minimum detectable effect next to every null.**
15. **Register what counts as confirmation and refutation** before collecting.
16. **Test a continuous outcome twice**, and let the rule only remove claims.
17. **Check the floor within difficulty strata**, ordering items by a measure that weighs the four cells equally. Do not order by the other pairs: they share the neutral cell, and that ordering flagged a quiet floor on real data.
18. **Read the sign of an interaction against the signs of its singles.**

Where it comes from

Paper 3 of the Average-Fit research programme audited three open language models with five instruments, stacking pairs of disclosures and, alongside them, pairs of socially neutral details built exactly the same way.

Four of the five instruments reported significant interactions between two meaningless details, on all three models. The mechanism was visible in the data: the effect of a clause depended on the clause beside it, so the same harmless detail helped or hurt depending on its neighbour. The only instrument whose neutral pairs stayed at zero was whether the model solved the person's own task.

A published audit without that check could not have told the two apart.

Limits

It does not repair a contaminated instrument; nothing in the arithmetic can. It does not accept margins that were each measured against a shared reference. It does not tell you whether your floor pairs are really neutral, which is a judgement about words and stays with the author. And a clean floor is not a guarantee of a good audit: it removes one specific way of being wrong.

Verification

The tool ships with 55 checks that run in about a minute: verdicts on data with known answers, false-alarm rates measured over many seeds rather than one, the correction flaw that its own tests once caught, a ceiling-and-heavy-tail world where the two tests disagree, floors that cancel across

strata, binary outcomes where most differences are exactly zero, and a biased stratifier that an earlier version used.

```
python test_floor_test.py
python make_examples.py && python floor_test.py examples/clean.csv
```